



Principles of Finance

Chapter 7

End-of-Chapter Questions

Time Value of Money I: Single Payment Value

Multiple Choice Questions

1. The most basic type of financial transaction involves _____.
 - A. an amount of money that is not invested
 - B. a series of equal installment amounts paid or received over a period of time
 - C. a simple, one-time amount of cash that can be either a receipt or a payment
 - D. None of the above

Solution: C

Section 7.1 LO 3 Easy

2. If a discount (or interest) rate has a positive value, then the future value of any amount deposited in an interest-bearing account will _____.
 - A. be less than the present value
 - B. be equal to the present value
 - C. be greater than the present value
 - D. decline over time

Solution: C

Section 7.3 LO 1 Easy

3. If the discount (or interest) rate used to calculate the present value of a future payment increases, the calculated present value will do which of the following?
 - A. Increase
 - B. Decrease
 - C. Remain the same
 - D. Increase as the period of time shortens

Solution: B

Section 7.3 LO 2 Easy

4. The discount rate that is required to equate a future payment of \$500 in three years to a present value of \$400 is _____.
 - A. 4.7%
 - B. 6.5%
 - C. 7.7%
 - D. 8.8%

Solution: C

Section 7.3 LO 3 Medium

5. If compounding periods increase in frequency and all else remains the same, the dollar values of any resulting future value calculations will _____.
- A. increase
 - B. remain the same
 - C. decrease
 - D. None of the above

Solution: A

Section 7.3 LO 1 Easy

Review Questions

1. All other things being the same, would you prefer a bank account that compounds interest quarterly or one that compounds interest semiannually?

Solution: Quarterly compounding is preferable because more frequent compounding can generate some exceptionally large differences in future values.

Section 7.2 LO 3 Easy

2. Briefly describe the relationship among consumer savings, purchasing power, and inflation.

Solution: The crux of TVM involves maintaining the present value of financial assets or increasing the value of financial assets at different points in the future when they may be needed to obtain goods and services. If a consumer's monetary assets grow at a greater rate than inflation over any period of time, then the consumer will realize an increase in their overall purchasing power. Conversely, if inflation exceeds savings or investment growth, then the consumer will lose purchasing power.

Section 7.4 LO 2 Difficult

3. Which of the following two options will give you the greatest future value: (A) an initial deposit of \$100 earning 20% per year, compounded annually and left to grow for 10 years, or (B) an initial deposit of \$75 earning 12% per year, compounded monthly and left to grow for 15 years?

Solution: Option A will provide the greatest future value. The FV of A is \$619.17, while the FV of B is only \$449.69.

Section 7.3 LO 1 Medium

4. If a savings account pays interest on a quarterly basis and you are performing future value calculations on deposited amounts, how can you calculate the rate?

Solution: To arrive at a quarterly rate to use in future value calculations, any annual period of time must be multiplied by 4 to accommodate the quarterly compounding periods. The quarterly rate that is arrived at through this method will then need to be multiplied by 4 to express it as an annual rate (if so desired).

Section 7.4 LO 4 Medium

5. Briefly describe the concept of future value within the context of the larger overall concept of the time value of money.

Solution: Future value refers to a greater value or a larger amount at some point in the future that is equivalent in value to a specific amount today, or the starting amount.

Section 7.4 LO 4 Medium

Problems

1. Find the future value of \$100 in five years at 5% interest.

Solution: \$127.63

Section 7.3 LO 1 Medium

2. Find the future value of \$1,800 in 3 years at 8% interest.

Solution: \$2,267.48

Section 7.3 LO 1 Medium

3. How much would you have to deposit now to have \$15,000 in eight years if interest is 7%?

Solution: \$,8730.14

Section 7.3 LO 2 Medium

4. What is the present value of \$5,000 that will be paid to you eight years from today at 8% interest?

Solution: \$2,701.34

Section 7.3 LO 2 Easy

5. How many years will it take a \$700 balance to grow into \$900 in an account earning 5%?

Solution: 5.15 years

Section 7.3 LO 2 Easy

6. If you borrow \$1,000 and pay back \$1,728 in three years, what annual rate of interest are you paying?

Solution: 20%

Section 7.3 LO 4Medium

7. How long will it take you to triple your money at 8%?

Solution: 14.27 years

Section 7.3 LO 5 Medium

8. A company's sales were \$250 million in 2019. If sales grow at 6% per year, how large will they be 10 years later, in 2029 (as expressed in millions)?

Solution: \$447.71 million

Section 7.3 LO 1 Medium

9. A US government bond in the amount of \$1,000 will mature in six years, has no coupon payments, and carries an interest rate of 8%. What is the value of this bond today?

Solution: \$630.17

Section 7.3 LO 2 Easy

10. You spend \$725.00 to purchase a \$1,000 bond that will have no coupon payments and matures in 12 years. What interest rate will you be earning on this bond?

Solution: 2.72%

Section 7.3 LO 3 Medium

11. At what interest rate would you be ambivalent about receiving either \$50,000 10 years from now or \$35,000 today?

Solution: 3.63%

Section 7.3 LO 2 Medium

12. Vance Corporation had earnings last year of \$3.25 per share. The company has experienced 8% annual growth over the last several years, and management expects that growth rate to continue. Based on this information, after how long will earnings per share double?

Solution: 9.0 years

Section 7.3 LO 5 Medium

13. What is the amount of total interest dollars earned on a \$5,000 deposit earning 6% for 20 years?

Solution: \$11,035.68

Section 7.3 LO 1 Medium

14. You have decided that you will sell your \$300,000 house when it appreciates in value to \$500,000. If houses are appreciating at an average annual rate of 5% in your neighborhood, for approximately how long will you be staying in your house?

Solution: 10.47 years

Section 7.3 LO 5 Medium

15. You just won some money in the lottery and would like to save a portion of it so that you will have \$50,000 to put a down payment on a house in five years. Your bank pays a 5% rate of interest. How much money will you have to set aside from the lottery winnings?

Solution: \$39,176.31

Section 7.3 LO 2 Medium

16. Bauer Bookstore sells books before they are published. Today, they offered the book *Journeys in Finance* for \$14.20, but the book will not be published for another two years. Upon publishing, the price of the book will be \$24.00. What is the discount rate Bauer Bookstore is offering its customers for this book?

Solution: 30%

Section 7.3 LO 3 Difficult

17. One of your professional goals is to one day earn a six-figure salary (\$100,000). You hope to accomplish this objective within the next 30 years. Salaries grow at 3.75% per year in your field of work. What beginning salary will you need in order to reach this 30-year goal?

Solution: \$33,140.33

Section 7.3 LO 32 Difficult

18. How much will \$25,000 grow to in five years at a 5% annual rate that is compounded quarterly?

Solution: \$32,050.93

Section 7.3 LO 1 Medium

19. If Eisenberg Industries revenues have increased from \$30 million to \$90 million over a 10-year period, what has been their annual rate of growth?

Solution: 11.61%

Section 7.3 LO 3 Medium

20. If you are scheduled to receive \$4,000 six years from today and the discount rate is 8.5%, what is the present value of this payment?

Solution: \$2,451.78

Section 7.3 LO 2 Medium

21. If you were to open a savings account that earns 3% interest and is compounded quarterly, what would be the total amount in your account after 10 years if you made an opening deposit of \$9,500?

Solution: \$12,809.31

Section 7.3 LO 1 Medium

22. You are considering four possible options for your new savings account. You plan to deposit \$13,000 and leave this amount in the account for 20 years with no additional deposits or withdrawals. All of these account options would earn 6% interest, but each one has a different compounding frequency, listed below. What would be the value of each account at the end of the 20-year period?

- | | |
|-----------------|-----------------------|
| a. Annually | Solution: \$41,692.76 |
| b. Semiannually | Solution: \$42,406.49 |
| c. Quarterly | Solution: \$42,778.62 |
| d. Monthly | Solution: \$43,032.66 |

Section 7.3 LO 1 Difficult

Chapter 7 Test Bank

Time Value of Money I: Single Payment Value

These questions and problems can all be solved in a classroom setting using a handheld financial calculator.

True or False Questions

1. Compounding a present value into the future at a zero-interest rate will generate a future value that is equal to its present value.

- A. True
- B. False

Solution: A. True

Section 7.2 LO 2 Easy

2. If one switches to a higher interest rate in discounting a future value, the result will be a smaller present value.

- A. True
- B. False

Solution: A. True

Section 7.3 LO 2 Easy

3. If there is no compounding within a year, a nominal interest rate will be equal to the effective interest rate.

- A. True
- B. False

Solution: A. True

Section 7.3 LO 3 Easy

4. The Fisher equation states that investors ignore inflation in assessing the rate of return that they get from an investment.

- A. True
- B. False

Solution: B. False

Section 7.4 LO 2 Easy

Multiple Choice Questions

1. If you invest \$3,200 today at 3 percent, how much will you have in 5 years?

- A. \$3,813
- B. \$3,710
- C. \$3,650
- D. \$2,870

Solution: B. Solve for FV, given $PV = 3,200$, $n = 5$, $r = 3$ percent

Section 7.3 LO 1 Moderate

2. How much would you have to invest today at 4.5 percent in order to have \$22,000 in 5 years?

- A. \$18,055.80
- B. \$20,900.92
- C. \$17,653.92
- D. \$19,305.90

Solution: C. Solve for PV, given $FV = 22,000$, $n = 5$, $r = 4.5$ percent

Section 7.3 LO 1 Easy

3. How much must you invest today at 5 percent if you want to have \$10,300 in 7 years?

- A. \$4,320
- B. \$6,930
- C. \$7,320
- D. \$8,302

Solution: C. Solve for PV, given $FV = 10,300$, $n = 7$, $r = 5$ percent

Section 7.3 LO 2 Moderate

4. What interest rate must your bank offer if you want your \$2,730 invested today to grow to \$3,470 in 3 years? Round to two places past the decimal point.

- A. 6.95 percent
- B. 7.52 percent
- C. 11.56 percent
- D. 8.32 percent

Solution: D. Solve for r , given $PV = 2,730$, $FV = 3,470$, $n = 3$

Section 7.3 LO 3 Moderate

5. How many years must you wait for your \$4,380 investment to grow to \$7,491.29 if you invest at 5 percent per year?

- A. 14
- B. 9
- C. 11
- D. 7

Solution: A. Solve for n , given $PV = 4,380$, $FV = 7,700$, $r = 5$ percent

Section 7.3 LO 5 Moderate

6. How long will it take for the population of Mudville, 5,205 individuals, to double if the population grows at 1.4 percent every year?
- A. 48 years
 - B. 72 years
 - C. 38 years
 - D. 50 years

Solution: D. Solve for n , given $PV = 5,205$, $FV = 10,410$ and $r = 1.4$ percent

Section 7.3 LO 5 Moderate

7. With a retirement plan target of having \$250,000 in 20 years, how much interest must you earn on your \$72,500 in order to achieve this goal?
- A. 10.85 percent
 - B. 8.62 percent
 - C. 7.50 percent
 - D. 6.38 percent

Solution: C. Solve for r , given $PV = 72,500$, $FV = 250,000$, $n = 20$

Section 7.4 LO 1 Moderate

Problems

1. What is the future value of \$1,750 in 3 years at an interest rate of 4 percent?

Solution: Solve for FV, given $PV = 1,750$, $n = 3$, and $r = .04$. \$1,968.51

Section 7.3 LO 1 Moderate

2. What is the future value of \$1,750 in 3 years at an interest rate of 5 percent?

Solution: Solve for FV, given $PV = 1,750$, $n = 3$, and $r = .05$. \$2,025.84

Section 7.3 LO 1 Moderate

3. What is the future value of \$1,750 in 3 years at an interest rate of 6 percent?

Solution: Solve for FV, given $PV = 1,750$, $n = 3$, and $r = .06$. \$2,084.28

Section 7.3 LO 1 Moderate

4. What is the present value of \$2,350 in 5 years at an interest rate of 3 percent?

Solution: Solve for PV, given $FV = 2,350$, $n = 5$, and $r = .03$. \$2,027

Section 7.3 LO 2 Moderate

5. What is the present value of \$2,350 in 5 years at an interest rate of 4 percent?

Solution: Solve for PV, given $FV = 2,350$, $n = 5$, and $r = .04$. \$1,931

Section 7.3 LO 2 Moderate

6. What is the present value of \$2,350 in 5 years at an interest rate of 5 percent?

Solution: Solve for PV, given $FV = 2,350$, $n = 5$, and $r = .05$. \$1,841

Section 7.3 LO 2 Moderate

7. If you invest \$2,700 for 20 years and it doubles in that time frame, what is the interest rate at which you made your investment?

Solution: Solve for r , given $PV = 2,700$, $n = 20$, and $FV =$ twice the PV. 3.53 percent

Section 7.3 LO 3 Difficult

8. If you invest \$2,700 for 20 years and it triples in that time frame, what is the interest rate at which you made your investment?

Solution: Solve for r , given $PV = 2,700$, $n = 20$, and $FV =$ three times the PV. 5.65 percent

Section 7.3 LO 3 Difficult

9. If you invest \$2,700 for 20 years and it quadruples in that time frame, what is the interest rate at which you made your investment?

Solution: Solve for r , given $PV = 2,700$, $n = 20$, and $FV =$ four times the PV. 7.18 percent

Section 7.3 LO 3 Difficult

10. How many years will it take for \$3,700 to double at an annual growth rate of 1.8 percent?

Solution: Solve for n , given $FV =$ double 3,700 and $r = .018$. 39 years

Section 7.3 LO 4 Difficult

11. How many years will it take for \$3,700 to triple at an annual growth rate of 1.8 percent?

Solution: Solve for n , given $FV =$ triple 3,700 and $r = .018$. 62 years

Section 7.3 LO 4 Difficult

12. How many years will it take for \$3,700 to quadruple at an annual growth rate of 1.8 percent?

Solution: Solve for n , given $FV =$ quadruple 3,700 and $r = .018$. 78 years

Section 7.3 LO 4 Difficult